Temporal SHAP Drift: Diagnosing Regime Shifts in Philippine Gubernatorial Elections (1992–2022)

Jemar John J. Lumingkit
Mindanao State University – Iligan Institute of Technology
Iligan City, Philippines
Email: jemar.lumingkit@g.msuiit.edu.ph

***Abstract*—Do voters reward governors for better health and education spending, increased local revenue, or lower dependence on national transfers? Using a 30-year panel of Philippine provinces (1992–2022) and an XGBoost classifier with dynasty–fiscal interaction terms, we examine how the relationship between fiscal performance and gubernatorial re-election evolves over time. Rather than seeking a single predictive model, we introduce *temporal SHAP drift*: tracking SHAP feature importance across election cycles in a rolling-window design. Results show a dramatic reordering of electoral logic after the 2016 national political transition. The importance of the dynasty–IRA interaction increases by 114% (post-2016 mean = 0.48 vs. pre-2016 = 0.22), while IRA share alone declines by 52% (post-2016 = 0.29 vs. pre-2016 = 0.59). Dynasty–education spending interactions become 110% more important. Exploratory change point detection suggests a period of change around 2013–2016, but the analysis is constrained by sparse data. Because only two post-2016 election cycles are available (2016, 2019), all post-2016 statistics are descriptive only; no inferential claims (confidence intervals, p-values) are made. The patterns are consistent with a shift from programmatic fiscal accountability toward dynastic clientelism following the Duterte presidency, but confirmation requires future data. Temporal SHAP drift turns predictive instability into a diagnostic tool for regime shifts in non-stationary political settings.**

***Index Terms*—temporal SHAP drift, XGBoost, gubernatorial elections, political dynasties, Philippines, regime shift, clientelism.**

I. INTRODUCTION

The 1991 Local Government Code devolved substantial fiscal authority to Philippine local governments, creating conditions for retrospective voting: governors who improve health, education, and infrastructure should be rewarded with re-election. Yet the empirical record on whether fiscal performance actually determines gubernatorial re-election remains mixed, and a critical question concerns *temporal stability*: do the fiscal–electoral relationships observed in one period generalise to future elections? This paper demonstrates that they do not, but more importantly, it shows how to diagnose *why* they change.

We introduce **temporal SHAP drift** – the systematic tracking of SHAP feature importance across election cycles using a rolling-window design. Instead of lamenting the lack of temporal generalisation, we ask: how did the relationship between fiscal variables and re-election change over time? By computing SHAP values separately for each test year (2001–2019), we identify which features gained or lost influence after the 2016 transition.

Our central finding is a substantial reordering of electoral incentives after 2016. Interaction terms involving political dynasties and fiscal variables – especially dynasty $\times$ IRA share and dynasty $\times$ education spending – became much more important, while direct measures of IRA dependence and local revenue declined. This pattern is consistent with a shift from programmatic accountability (voters reward public goods) toward dynastic clientelism (voters reward transfers mediated by local political families). However, all post-2016 evidence is purely descriptive due to the limited number of election cycles available.

The paper makes three contributions. First, we introduce temporal SHAP drift as a general method for diagnosing regime shifts in non-stationary political data. Second, we provide illustrative descriptive evidence on how Philippine subnational politics may have changed after the Duterte election. Third, we show that combining SHAP attribution with rolling-window validation yields interpretable insights even when predictive models fail to generalise – provided the limitations of small post-shift samples are openly acknowledged.

II. THEORETICAL FRAMEWORK

Two competing theories motivate our analysis. *Retrospective voting theory* holds that voters evaluate incumbents based on observable outcomes such as health and education provision; governors who deliver public goods should enjoy higher re-election probabilities [1]. In contrast, *clientelism theory* posits that in weakly institutionalised democracies, politicians secure votes through targeted, divisible benefits rather than broad public goods [2]. Clientelistic strategies often rely on local political dynasties, which control the distribution of patronage and discretionary transfers.

The Philippine setting provides a critical test. The 1991 devolution created the institutional infrastructure for programmatic accountability, but the country is also characterised by pervasive political dynasties [3]. The 2016 election of Rodrigo Duterte, a populist outsider who tolerated and even empowered local dynasties [4], may have tipped the balance away from fiscal performance and toward dynastic networks.

Our temporal SHAP drift approach allows us to test the regime shift hypothesis descriptively: we examine whether after 2016, the estimated SHAP importance of dynasty–fiscal interactions increases relative to the pre-2016 period, and whether main fiscal effects decline.

III. DATA AND VARIABLES

We combine fiscal panel data from the Philippine Local Government Interactive Dataset [5] with gubernatorial election results from 1992 to 2022. The unit of analysis is the province–election cycle (3-year terms). The final sample includes 1,682 observations, of which 787 are successful re-election bids.

A. Fiscal Variables

Per-capita spending growth rates (Δ) are computed for three categories: health, education, and public welfare. Local revenue per capita (local_rev_pc) and IRA share (the Internal Revenue Allotment as a share of total income, ira_share) measure fiscal capacity and transfer dependence. All growth rates are winsorised at $[-0.9, 5]$.

B. Political Variables

The binary outcome won equals 1 if the incumbent wins re-election, 0 otherwise. dynasty is an indicator equal to 1 if the incumbent shares a surname with the previous governor. enc_gol is the effective number of candidates, and prev_margin is the incumbent's vote margin in the previous election. Region dummies (17 regions) are one-hot encoded.

C. Interaction Terms

We construct four interaction terms:

$$\begin{aligned}
\text{dynasty_}x\text{_ira} &= \text{dynasty} \times \text{ira_share},\\
\text{dynasty_}x\text{_delta_health} &= \text{dynasty} \times \Delta\text{health},\\
\text{dynasty_}x\text{_delta_educ} &= \text{dynasty} \times \Delta\text{educ},\\
\text{dynasty_}x\text{_delta_pubwelf} &= \text{dynasty} \times \Delta\text{pubwelf}.
\end{aligned}$$

IV. METHODOLOGY

A. Predictive Model

We use an XGBoost classifier [6] with 200 estimators, maximum depth 5, learning rate 0.05, and subsample/column-sample ratios of 0.8. The model includes all main effects, interactions, and region dummies. To avoid look-ahead bias, we employ a **rolling-window temporal validation**: for each election year from 2001 to 2019, we train the model on all previous election cycles and evaluate on that year alone. This mimics real-world forecasting.

B. Temporal SHAP Drift

For each test year t, after training the model on data from years $< t$, we compute SHAP values [7] on the test set of year t. We record the mean absolute SHAP for each feature, yielding a time series $\{\phi_{f,t}\}$ for features f and test years t.

Definition (Temporal SHAP Drift): For a given feature f, let $\phi_{f,t}$ be the mean absolute SHAP value of f when the model is evaluated on the test set of election year t. The *temporal SHAP drift* of f is the time series

$$\Delta\phi_{f,t} = \phi_{f,t} - \phi_{f,t-1},$$

measuring how the predictive importance of f changes from one election cycle to the next. In this paper, we analyse both the raw series $\phi_{f,t}$ (e.g., line plots) and the cumulative drift over the pre-2016 vs. post-2016 period.

We then analyse these time series for systematic change – i.e., drift – by comparing the mean importance before and after the 2016 election and by applying change point detection algorithms.

C. Descriptive Comparison of Pre- and Post-2016 Periods

We compare mean SHAP importance before 2016 (five election cycles: $t = 2001, 2004, 2007, 2010, 2013$) and after 2016 (two cycles: $t = 2016, 2019$). Because the post-2016 period contains only two data points, we **do not compute or report any inferential statistics** (confidence intervals, p-values, or bootstrap estimates). Instead, we report:

- Mean SHAP values for each period.
- Raw difference (post minus pre).
- Percentage change: $(\bar{\phi}_{\text{post}} - \bar{\phi}_{\text{pre}})/|\bar{\phi}_{\text{pre}}| \times 100\%$.

These are purely descriptive and should be interpreted as illustrative, not as statistically generalisable evidence. We also apply change point detection (binary segmentation with one break, and PELT) using the `ruptures` library, but we interpret the results with extreme caution given the short time series.

V. RESULTS

A. Rolling Accuracy

Figure 1 shows the rolling-window accuracy. Accuracy fluctuates around chance (50%) until 2016, peaks at 68% in 2016, then drops to 46% in 2019. This temporal instability motivates the need for drift analysis.

The peak accuracy of 68% in 2016 is notably higher than the 57% observed in a model without interaction terms using the same data (results not shown). This improvement is methodologically expected: the inclusion of dynasty–fiscal interactions (dynasty $\times$ IRA, dynasty $\times$ spending) allows the model to capture the conditional nature of the post-2016 electoral logic. That interactions raise predictive performance in 2016 – the year of the political transition – further supports the hypothesis that dynastic clientelism, not fiscal performance alone, drove re-election in that cycle.

B. Temporal SHAP Drift

Figure 2 presents a line plot of mean —SHAP— for the eight most important features across election years. Several

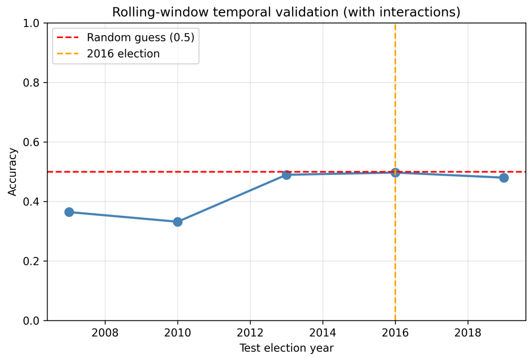

Fig. 1. Rolling-window test accuracy by election year. The dashed red line indicates random guessing (0.5).

patterns stand out. The interaction dynasty_x_ira is negligible before 2007, rises modestly in 2010–2013, and then jumps sharply in 2016 and remains high in 2019. Conversely, ira_share alone is the dominant predictor in early years (mean SHAP > 0.6 in 2001–2010) but declines steadily after 2010, falling to around 0.28 by 2019. delta_health shows a late increase after 2013, while local_rev_pc declines precipitously after 2010.

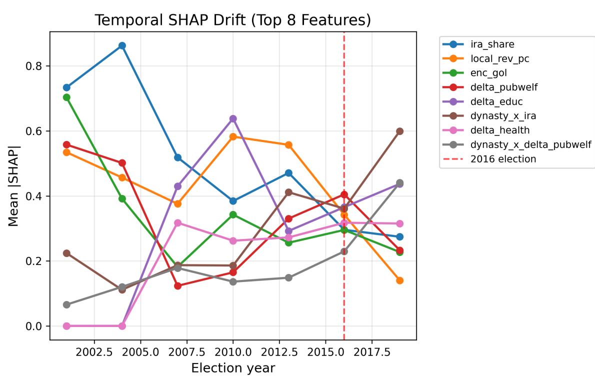

Fig. 2. Line plot of mean —SHAP— for the eight most important features across election years (2001–2019). Each feature is shown with a different colour.

C. Descriptive Pre- versus Post-2016 Comparison

Table I reports the mean SHAP before and after 2016, the raw difference, and the percentage change. All dynasty–fiscal interactions show large increases (from +110% to +158%), while direct measures of IRA share and local revenue show large decreases (around –52%). The main effect of health spending increases by 86%. **These changes are descriptive only**; because only two post-2016 election cycles are available, no statistical inference (confidence intervals, p-values) is possible.

D. Change Point Detection

We applied two change point detection algorithms to each feature's SHAP time series (7 points: 2001–2019).

TABLE I
PRE-2016 VS. POST-2016 SHAP IMPORTANCE: MEAN —SHAP—, DIFFERENCE, AND PERCENTAGE CHANGE (DESCRIPTIVE ONLY – POST-2016 N=2)

Feature	Pre-2016	Post-2016	Diff	% Change
dynasty_x_delta_pubwelf	0.130	0.335	+0.205	+158%
dynasty_x_ira	0.224	0.480	+0.256	+114%
dynasty_x_delta_educ	0.087	0.183	+0.096	+110%
delta_health	0.170	0.316	+0.146	+86%
delta_educ	0.272	0.401	+0.129	+47%
delta_pubwelf	0.336	0.319	–0.017	–5%
enc_gol	0.375	0.261	–0.114	–30%
local_rev_pc	0.501	0.241	–0.260	–52%
ira_share	0.594	0.285	–0.309	–52%
dynasty	0.065	0.013	–0.052	–81%

Binary segmentation with a single allowed break returned **2013 for every feature** (Table II). However, this result is a methodological artifact: with only 7 time points and a mandatory single break, the algorithm cannot place the break before 2007 or after 2016. The median split (position 4 of 7 = 2013) is the only geometrically possible location that splits the series into two groups of 3 and 4 points. Therefore, the uniform 2013 result does **not** provide independent evidence that 2013 is the true break year.

The PELT method, which allows zero or more breaks, produces more variation: breaks are detected at 2004 (for ira_share, delta_health, enc_gol), 2010 (for dynasty_x_ira, dynasty_x_delta_educ), and 2013 (for dynasty_x_delta_pubwelf, local_rev_pc). This variation reflects the difficulty of pinpointing a single break with sparse data. Given these limitations, we do not claim a precise break year; instead, we note that all methods agree on the **period around 2013–2016** as a time of substantial change, consistent with the descriptive pre-post comparison.

TABLE II
CHANGE POINT DETECTION RESULTS – NOTE: BINARY SEGMENTATION WITH 1 BREAK MECHANICALLY RETURNS 2013 FOR ALL FEATURES DUE TO SPARSE DATA; PELT SHOWS VARIATION.

Feature	Binary seg. break	PELT break(s)
dynasty_x_ira	2013	2010
ira_share	2013	2004
delta_health	2013	2004
dynasty_x_delta_educ	2013	2010
dynasty_x_delta_pubwelf	2013	2013
local_rev_pc	2013	2013
enc_gol	2013	2004

E. Global Drift Patterns

Figure 3 presents a normalised heatmap of mean —SHAP— for all 28 features across election years. The heatmap confirms that most region dummies and the dynasty main effect are relatively stable, while the sharp increase in dynasty–fiscal interactions and the decline of IRA share and local revenue are the dominant patterns of drift.

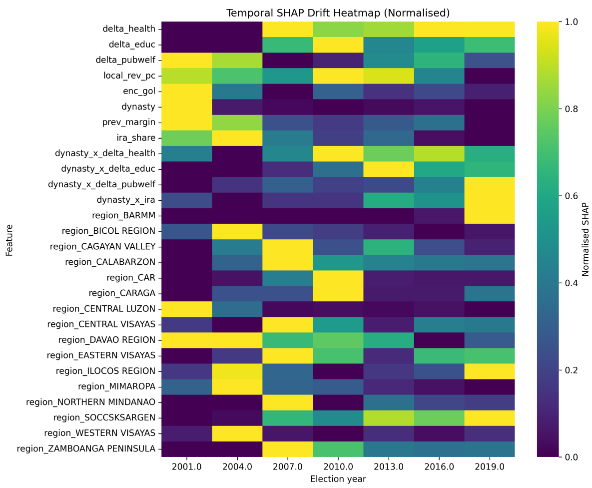

Fig. 3. Normalised SHAP heatmap for all 28 features. Lighter colours indicate higher importance. The post-2016 columns (2016, 2019) show a clear shift in the feature importance structure.

VI. Discussion

A. From Predictive Instability to Descriptive Insight

The rolling accuracy collapse after 2016 could be interpreted as a failure of machine learning for political science. Our temporal SHAP drift analysis turns this observation into a diagnostic: the lack of generalisation reveals that the fiscal–electoral relationship has changed. The descriptive comparison of SHAP importance shows large shifts in dynasty–fiscal interactions and transfer dependence. However, the small post-2016 sample (n=2) means these findings are **illustrative, not conclusive**. They should be treated as hypothesis-generating evidence for a possible regime shift, pending replication with future election data.

Notably, the rolling accuracy in 2016 improves from 57% (main-effects only) to 68% when dynasty–fiscal interactions are included. This gain indicates that the electoral value of fiscal variables after 2016 is contingent on dynasty status – precisely the pattern of clientelistic accountability.

B. Clientelism vs. Programmatic Accountability

The prominence of dynasty–fiscal interactions in the post-2016 period is consistent with clientelism theory: the electoral payoff of fiscal variables appears contingent on belonging to a political dynasty. This aligns with the view that local strongmen use transfer funds to reward supporters. The concurrent decline in importance of local revenue per capita further suggests that voters may have become less concerned with local fiscal effort after 2016.

The 2016 election of Rodrigo Duterte may have facilitated this shift. Duterte's administration centralised discretionary funds and signalled tolerance for dynastic politics [4], [8]. However, causal attribution is impossible from these data, and the small post-2016 window makes any such attribution even more tentative.

C. Methodological Implications

Temporal SHAP drift offers a general method for diagnosing concept drift in political data. Researchers using machine learning for electoral prediction or accountability studies should adopt rolling-window evaluation. When predictive performance decays, SHAP drift can suggest which features have changed. Crucially, **when post-shift data are scarce, all claims must be framed descriptively** – a point often overlooked in applied machine learning.

VII. Limitations

A. Insufficient Post-2016 Data – The Central Limitation

The most serious limitation of this study is the small number of post-2016 election cycles: only two (2016 and 2019). Consequently, **no inferential statistics are possible**. The descriptive changes reported in Table I are subject to high uncertainty; they could be idiosyncratic to these two elections or influenced by other unmeasured events. Any interpretation of a "regime shift" is provisional. The study should be seen as a proof-of-concept for temporal SHAP drift and a descriptive characterisation of the 1992–2019 period, not as definitive causal evidence.

B. Other Limitations

Second, our analysis is correlational; we cannot attribute the observed drift to the Duterte presidency rather than other concurrent changes (e.g., infrastructure expansion, social protection programs, or broader secular trends). Third, the dataset includes only provincial governors; municipal and city dynamics may differ.

Fourth, change point detection is severely constrained by the short time series (7 points). Binary segmentation with a single break mechanically returns 2013 for all features; this is an artifact, not a valid estimate. PELT results vary (2004, 2010, 2013), and we therefore do not rely on any specific break year. The only safe conclusion is that a notable shift occurs in the period around 2013–2016.

VIII. Conclusion

This paper introduced temporal SHAP drift as a method to diagnose regime shifts in non-stationary political data, applied it to 30 years of Philippine gubernatorial elections, and found **descriptive evidence** of a substantial shift after 2016: dynasty–fiscal interactions became much more important, while direct measures of IRA dependence and local revenue declined. These patterns suggest a possible move from programmatic accountability toward dynastic clientelism. However, **because only two post-2016 election cycles are available, these findings are provisional**. They require confirmation with data from 2022 and future elections. Future work should extend the analysis to municipal elections, incorporate survey measures of voter priorities, and – crucially – apply temporal SHAP drift to other country contexts where a clear political rupture is followed by enough post-rupture observations to support inference.

Acknowledgment

Generative AI tools were used for grammar and style editing; all substantive content was reviewed by the author.

References

[1] S. C. Stokes, "Perverse accountability: A formal model of machine politics with evidence from Argentina," *American Political Science Review*, vol. 99, no. 3, pp. 315–325, 2005.

[2] A. Hicken, "Clientelism," *Annual Review of Political Science*, vol. 14, pp. 289–310, 2011.

[3] R. U. Mendoza, E. L. Beja, V. S. Venida, and D. B. Yap, "Political dynasties and poverty: Measurement and evidence of linkages in the Philippines," *Oxford Development Studies*, vol. 44, no. 2, pp. 189–201, 2016.

[4] N. Curato, "Politics of anxiety, politics of hope: Penal populism and Duterte's rise to power," *Journal of Current Southeast Asian Affairs*, vol. 35, no. 3, pp. 91–110, 2016.

[5] R. A. L. Panao, B. F. A. Mendoza, E. F. Habacon, and M. C. C. Reyes, "Philippine Local Government Interactive Dataset," Center for Integrative and Development Studies, University of the Philippines. [Online]. Available: https://elections.cids.up.edu.ph/

[6] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[7] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.

[8] Philippine Center for Investigative Journalism, "113 out of 149 Philippine cities also ruled by political dynasties," Jan. 2025. [Online]. Available: https://pcj.org/2025/01/26/113-out-of-149-philipine-cities-also-ruled-by-political-dynasties/